

Personal Cognitive Infrastructure

The Engineering Problem and Research Programme of Personal Cognitive Computing

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Published under: Cassian Bey

Status: Foundational Research Paper

Version: v1.0

Publication date: 7 August 2026

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Relationship to the Canonical Documents

This paper accompanies two foundational publications:

1. *Personal Cognitive Computing - The Next Chapter in the History of Computing*
2. *Personal Cognitive Engineering - Mission, Engineering Methodology & Reference Architecture*

Each document fulfils a distinct role within the emerging discipline.

The first establishes the historical thesis: that computing is entering a new phase in which the primary computing environment shifts from organisations and cloud platforms toward the individual. It argues that this transition is not merely technological, but represents a structural evolution in the relationship between people and computing systems.

The second translates that thesis into engineering: it defines the mission of Personal Cognitive Computing, introduces the engineering methodology through which the discipline should mature, presents a reference architecture, and describes Nivel as an early reference implementation through which architectural ideas can be explored, validated and refined.

This third paper serves a different purpose: rather than arguing *why* Personal Cognitive Computing is necessary, or describing *how* a reference architecture may be organised, it asks what remains fundamentally unsolved. It establishes the research domains, methodological principles and conceptual basis required for the future development of shared evaluation criteria for Personal Cognitive Computing architectures, systems and implementations; its objective is therefore neither architectural nor historical; its objective is scientific.

A discipline matures not merely by proposing architectures, but by identifying the underlying questions that those architectures attempt to answer. Those questions should remain meaningful even as today's models, hardware platforms and software frameworks inevitably change.

Accordingly, this paper seeks to establish a durable research programme whose relevance extends beyond any particular implementation.

Taken together, the three foundational papers provide complementary perspectives on the same field.

- **Historical Foundation:** Why Personal Cognitive Computing represents a distinct transition in the history of computing.
- **Mission and Reference Architecture:** What Personal Cognitive Computing is, the principles that guide it, and one possible architectural reference.
- **Scientific Foundation:** Which engineering problems remain unresolved, which research domains follow from them, and how shared evaluation criteria may progressively be developed.

No single document attempts to fulfil all three roles; together they establish the constitutional, architectural and scientific foundations of the discipline.

Reading Guide

This paper is organised around three fundamental questions.

1. What is the unresolved engineering problem?

Personal Cognitive Computing begins with the observation that contemporary language models have demonstrated remarkable reasoning capabilities, yet they remain embedded within computing environments that were not designed around persistent, user-owned cognition.

The central engineering challenge is therefore not simply to build better models.

It is to determine how computational reasoning can become a trustworthy, inspectable and durable component of an individual's own computing environment while preserving personal ownership, continuity and agency.

This paper treats that challenge as the defining engineering problem of the discipline.

2. What enduring research questions follow from that problem?

Engineering disciplines are defined less by their current implementations than by the questions they continue to investigate.

Some questions concern architecture.

Others concern retrieval, memory, transparency, evaluation, interaction, governance or long-term sustainability.

Many of these questions will remain open irrespective of future advances in language models or hardware.

This paper identifies those enduring questions and proposes them as a long-term research agenda for Personal Cognitive Computing.

3. How can shared evaluation criteria be developed?

Future systems will inevitably differ in architecture, implementation strategy and technical capability.

Without shared evaluation principles, however, it becomes difficult to distinguish cumulative progress from architectural divergence, incompatible assumptions or isolated technological demonstrations.

This paper therefore establishes the research domains, methodological principles and conceptual basis required for the future development of shared evaluation criteria for Personal Cognitive Computing architectures, infrastructures and implementations.

Such criteria should ultimately support meaningful comparison, guide engineering decisions and enable scientific and technical findings to accumulate across different systems, experiments and implementation strategies. Their development must itself remain an explicit subject of research rather than being prematurely presented as a completed framework.

This paper should not be understood as a product roadmap, an implementation specification or a prediction of future technologies.

- It is **not** a development plan for Nivel, nor does it prescribe a single implementation strategy.
- It does **not** argue that all computing should be performed locally, nor does it benchmark contemporary language models or compare competing commercial systems.
- It does **not** define a completed evaluation framework for Personal Cognitive Computing.
- Likewise, it does **not** claim that Personal Cognitive Computing already constitutes a mature engineering discipline.

Instead, it recognises that the field remains in its formative stage.

Its purpose is to identify the enduring engineering problems that must be investigated, organise the principal research domains that follow from those problems, and establish the methodological foundation from which future research questions, experiments and evaluation criteria may be developed.

In that sense, this document forms the bridge between the constitutional and architectural foundations established by the preceding papers and the practical research required to transform Personal Cognitive Computing from an emerging concept into a mature and scientifically grounded discipline.

Executive Summary

Artificial intelligence is becoming cognitive infrastructure. Yet the dominant architecture through which it is delivered places models, memory, identity, storage, retrieval and execution under provider control. Open-weight models and local inference partially reopen the possibility of personal ownership, but neither model portability nor local execution is sufficient. Personal Cognitive Computing therefore addresses a broader engineering problem: how to construct durable, affordable, inspectable and user-governed cognitive environments on infrastructure ordinary people can realistically own and operate.

The discipline investigates how information is stored, scoped, retrieved, assembled as evidence, submitted to reasoning systems, preserved as memory and governed across time. It evaluates systems not by model scale or commercial reach, but by ownership, continuity, transparency, evidence traceability, user authority, affordability and architectural independence.

Nivel is the current reference implementation through which these questions are tested, but it does not define the discipline. The research programme must remain implementation-independent, technologically neutral and empirically grounded.

Foundational Thesis

Personal Cognitive Computing is the engineering discipline concerned with the construction, governance and preservation of user-controlled cognitive environments.

This proposition follows directly from the historical and architectural foundations established in the preceding canonical papers.

The historical paper argues that computing has entered a new transition. As language models transform computers from deterministic tools into cognitive systems, the defining question is no longer merely what these systems are capable of, but who ultimately governs the architecture through which those capabilities are exercised. The transition is therefore not solely technological. It concerns the relationship between human beings and the cognitive systems that increasingly assist, organise and influence their thinking.

The architectural paper responds to this transition by introducing the Personal Cognitive Environment as the enduring foundation beneath models, applications and external services. It argues that while reasoning models will continue to evolve, the user's own cognitive environment must remain stable, inspectable and under explicit personal control. Models become replaceable components; the individual's cognitive infrastructure becomes the enduring centre of gravity.

Taken together, these propositions naturally define a new engineering discipline.

Personal Cognitive Computing is not the study of language models, nor is it the design of conversational applications. Its concern is the engineering of complete cognitive

environments that remain coherent, trustworthy and durable throughout continuous technological change. It encompasses the principles, architectures, methodologies and engineering practices required to ensure that an individual's cognitive environment remains governed by the individual rather than by the transient characteristics of particular models, vendors or computing platforms.

Within this discipline, **Personal Cognitive Infrastructure** constitutes the technical and organisational foundation through which these objectives are realised. It comprises the architectural structures, shared capabilities, governance mechanisms and operational principles that allow a Personal Cognitive Environment to remain persistent, inspectable and under explicit user control. Infrastructure, in this context, extends beyond software architecture. It includes the enduring mechanisms through which cognitive continuity, transparency, ownership and long-term preservation are maintained despite continual advances in artificial intelligence and computing technology.

The discipline ultimately serves a broader objective.

Personal Cognitive Sovereignty describes the human and societal condition that Personal Cognitive Infrastructure seeks to preserve. It is the condition in which individuals retain meaningful authority over the systems that increasingly participate in their reasoning, memory and decision-making. Sovereignty is therefore not itself an engineering discipline, nor is it merely a political aspiration. It is the intended outcome of sound engineering: the preservation of human agency through architectures designed to place the individual, rather than external institutions or service providers, at the centre of the cognitive computing environment.

These three concepts occupy distinct but complementary roles.

- Personal Cognitive Computing defines the engineering discipline.
- Personal Cognitive Infrastructure provides the architectural and governance foundation through which the discipline is implemented.
- Personal Cognitive Sovereignty expresses the enduring human objective that this infrastructure exists to protect.

Within this emerging discipline, **Nivel** serves as the present reference implementation. It is not proposed as the discipline itself, nor as its final architecture, but as an evolving engineering vehicle through which foundational principles, architectural concepts and research hypotheses can be explored, validated and refined. As with any reference implementation, its purpose is to contribute to the maturation of the discipline rather than to define its boundaries.

Chapter I - The Engineering Problem of Personal Cognitive Computing

1. Purpose of the Chapter

Every engineering discipline begins by defining the problem it intends to solve.

This principle is particularly important for Personal Cognitive Computing because the field has emerged during a period of extraordinary progress in artificial intelligence. The rapid improvement of language models has understandably drawn attention toward models themselves—their capabilities, benchmarks, parameter counts and deployment strategies. Yet none of these developments, significant as they are, fully explains why a distinct engineering discipline should exist.

If Personal Cognitive Computing were simply the practice of running language models locally, no new discipline would be required. It would represent a deployment preference rather than an architectural transition.

Likewise, if the objective were merely to increase privacy, release software as open source, improve retrieval quality or build more capable personal assistants, existing disciplines would already provide the necessary engineering frameworks.

Personal Cognitive Computing therefore begins from a different observation.

The emergence of increasingly capable reasoning systems has created a new architectural condition that existing computing paradigms were never designed to address.

As cognitive capabilities become integrated into everyday computing, individuals no longer interact with isolated software applications alone. They begin to construct persistent cognitive environments composed of personal knowledge, long-term context, documents, memories, preferences, workflows, reasoning services and increasingly interchangeable models. These environments evolve over years rather than sessions, survive the replacement of individual software components and become progressively more central to how individuals organise, understand and extend their own thinking.

The unresolved engineering problem is not how to execute a model locally.

The unresolved engineering problem is how to construct, govern and preserve such cognitive environments as durable, user-controlled systems.

This distinction is fundamental.

Personal Cognitive Computing is not synonymous with local artificial intelligence.

A language model may execute entirely on personal hardware while remaining dependent upon opaque software architectures, fragmented data management, transient conversational interfaces or vendor-specific ecosystems. Local execution alone provides no guarantee of continuity, inspectability, portability or long-term ownership.

Nor is Personal Cognitive Computing equivalent to private artificial intelligence.

Privacy protects information from unauthorized observation. It does not determine how cognitive systems are organised, how reasoning is governed, how evidence is managed or whether the resulting environment remains durable across technological change.

Neither is it synonymous with open-source software.

Open source improves transparency of implementation, but the availability of source code alone does not establish coherent cognitive architectures, durable governance models or persistent user-controlled environments.

Likewise, Personal Cognitive Computing cannot be reduced to retrieval-augmented generation, personal knowledge management or conversational interfaces.

- Each addresses an important technical capability.
- None addresses the architecture within which those capabilities must coexist over decades of technological evolution.

Nor should Personal Cognitive Computing be understood as an attempt to recreate cloud artificial intelligence on smaller hardware.

The objective is not to replicate cloud services inside the home.

It is to engineer an entirely different architectural relationship in which the individual's own cognitive environment becomes the enduring centre of the system, while models, services and external capabilities remain replaceable components within that environment.

Finally, Personal Cognitive Computing should not be interpreted as an ideological rejection of external services or distributed computing.

Future cognitive environments will almost certainly interact with remote models, specialised services, collaborative infrastructures and shared computational resources whenever such interactions are beneficial and explicitly chosen by the user.

The defining question is therefore not **where** computation occurs.

The defining question is **where architectural authority resides**.

- Can the individual's cognitive environment remain coherent, inspectable and under explicit user governance while making use of continuously changing models and services?
- Can continuity be preserved as models improve, providers disappear, hardware evolves and software ecosystems are replaced?
- Can the architecture itself ensure that personal cognition remains organised around the individual rather than around the commercial or technical assumptions of any particular platform?

These questions define the engineering problem of Personal Cognitive Computing.

They remain unresolved even in a future where every state-of-the-art language model can be downloaded, executed locally and inspected in full.

The challenge is no longer obtaining cognitive capability.

The challenge is engineering the enduring architecture within which that capability becomes a trustworthy, persistent and user-governed component of an individual's own cognitive environment.

2. From Computation to Cognitive Infrastructure

2.1 The Changing Role of Computing

The emergence of Personal Cognitive Computing cannot be understood solely through advances in artificial intelligence. It reflects a broader transition in the historical role of computing itself.

The earliest generations of computing systems were developed to automate calculation. Their value lay in performing arithmetic operations with speed, consistency and accuracy beyond human capability. As computing matured, its role expanded. Computers became systems for storing, organising and transmitting information. They evolved into the infrastructure through which institutions managed records, communication and increasingly complex digital processes.

Today a further transition is becoming visible. Computing systems no longer serve merely as instruments for calculation or repositories of information. They increasingly participate in activities that shape how knowledge is produced, interpreted and applied. Language models assist with research, summarise documents, generate explanations, organise evidence, draft correspondence, support planning and contribute to decision-making processes. They participate in the practical work of reasoning itself.

The significance of this development is frequently misunderstood.

The defining characteristic of the transition is not that machines have become human, nor that artificial intelligence has achieved human cognition. Such claims confuse capability with function.

The more consequential observation is architectural.

Computational systems increasingly participate in the processes through which human beings interpret information, construct understanding and make decisions. They become participants within human cognitive workflows rather than merely tools that execute predefined operations.

As a consequence, the engineering questions surrounding these systems also change.

- When software merely automates calculation, engineering focuses primarily on correctness, performance and reliability.
- When software becomes part of an individual's ongoing cognitive environment, additional concerns inevitably emerge. Continuity, inspectability, provenance, governance and long-term preservation become architectural properties rather than optional features.

The transition from computational systems to cognitive systems therefore demands a corresponding transition in engineering.

2.2 Why Cognitive Infrastructure Requires Different Engineering

Traditional software systems are generally designed around tasks.

A word processor edits documents. A spreadsheet performs calculations. An accounting system manages financial records. Although these applications may become important, they remain functionally bounded. They can usually be replaced without fundamentally disrupting the continuity of an individual's intellectual life.

Cognitive infrastructure occupies a different position.

Over time, it may accumulate the documents through which a person develops ideas, the correspondence that records professional relationships, the instructions that govern interaction with computational systems, the preferences that shape recurring behaviour and the research histories that explain how conclusions were reached. It may preserve evidential relationships between sources, maintain evolving interpretations of complex subjects, support recurring work processes and retain institutional knowledge that extends across projects, organisations and years of experience.

None of these elements derives its significance from isolation, their value emerges through continuity.

Together they form an evolving body of knowledge that reflects not merely what an individual knows, but how that knowledge has been acquired, organised, interpreted and refined over time.

Consequently, the failure of such infrastructure carries consequences fundamentally different from the failure of ordinary software.

- The loss of a text editor is inconvenient.
- The loss of a cognitive environment may sever the relationships between evidence, interpretation and decision that have accumulated over years. It may render previous reasoning difficult to reconstruct, obscure the provenance of conclusions or dissolve the organisational structures upon which future work depends.

Likewise, enclosure within proprietary ecosystems presents more than a commercial concern.

If cognitive continuity becomes inseparable from a particular vendor, service or technological platform, the individual gradually loses practical authority over the environment through which their own intellectual work is conducted. Even where data remains technically accessible, the surrounding structures that give that data meaning may become difficult to preserve or reproduce.

Engineering for cognitive infrastructure must therefore address properties that extend beyond functional capability.

It must preserve continuity across technological change, maintain transparent relationships between evidence and conclusions, support explicit governance by the user and ensure that accumulated cognitive assets remain durable independently of the particular models, interfaces or services employed at any given moment.

These requirements arise not from ideological preference but from the changing role that computing now occupies within human intellectual activity.

2.3 The Primary Engineering Object

The central engineering object of Personal Cognitive Computing is not the language model: language models are powerful reasoning components, but they remain components.

Their capabilities will continue to improve. Their architectures will evolve. They will be replaced, specialised and superseded by future generations of cognitive systems. Designing an engineering discipline around any particular model would therefore anchor the discipline to one of its most rapidly changing elements.

Personal Cognitive Computing instead places its architectural foundation elsewhere.

Its primary engineering object is the **Personal Cognitive Environment**: the durable body of user-controlled information, instructions, histories, evidence, relationships and derived knowledge upon which cognitive applications operate. This environment constitutes the enduring centre of the architecture.

It persists beyond individual reasoning sessions, beyond software releases and beyond successive generations of artificial intelligence. It provides the continuity through which cognitive work acquires long-term coherence, allowing new reasoning systems to build upon existing knowledge without becoming the owners of that knowledge.

Within this architectural perspective, language models, user interfaces and external services assume a different role.

They become replaceable participants surrounding the Personal Cognitive Environment rather than defining it. Their purpose is to assist the environment, not to contain it.

This principle follows directly from the reference architecture established in the companion paper. Layer 0 establishes the Personal Cognitive Environment as the permanent foundation upon which all higher architectural layers depend. Every subsequent component derives its purpose from supporting, protecting or extending that environment rather than replacing it.

This distinction is more than an architectural preference, it establishes a durable engineering principle.

As reasoning models continue to advance, Personal Cognitive Computing does not seek permanence in rapidly changing computational capabilities. It seeks permanence in the individual's own cognitive environment, ensuring that technological progress strengthens rather than displaces the continuity of human knowledge, reasoning and intellectual autonomy.

3. The Structural Problem

3.1 Use Without Ownership

The emergence of powerful cognitive services has dramatically lowered the barrier to advanced computational reasoning. Today, individuals can access capabilities that only a few years ago required specialised research laboratories or substantial computational resources. This accessibility represents an important technological achievement.

Accessibility, however, should not be confused with ownership.

An individual may rely upon a cognitive system every day, entrust it with research, planning, correspondence, professional work and personal knowledge, yet possess remarkably little authority over the architecture upon which that relationship depends.

The distinction is subtle but fundamental.

A user may accumulate years of interaction without owning the memory through which continuity is established. The retrieval processes that determine which information is presented may remain invisible. Identity, authentication and access may be governed entirely by external systems. Operating histories may not be independently preserved. Model configurations may change without the user's knowledge. The processes through which evidence is selected, prioritised or excluded may remain inaccessible for inspection. Even continued access to the accumulated cognitive environment may ultimately depend upon policies established by organisations entirely outside the user's control.

From the perspective of conventional software engineering, many of these arrangements appear entirely reasonable. They simplify deployment, reduce operational complexity and enable rapid innovation.

From the perspective of Personal Cognitive Computing, however, they reveal a deeper architectural condition.

The problem is not that users lack powerful cognitive capabilities.

The problem is that those capabilities may exist within environments over which users possess only limited structural authority.

Accordingly, the central concern is not simply privacy.

- Privacy governs who may observe information.
- Ownership governs who determines the architecture within which that information acquires meaning, continuity and operational value.

These questions are related, but they are not identical.

A system may provide excellent privacy protections while remaining structurally dependent upon infrastructures that the user cannot inspect, preserve, migrate or govern. Conversely, architectural ownership concerns the enduring capacity to retain authority over one's

cognitive environment independently of particular technologies, service providers or commercial arrangements.

The engineering challenge therefore extends beyond protecting information.

It concerns preserving authority over the environment through which that information becomes knowledge.

3.2 Cognitive Recentralization

The history of computing has often been characterised as a gradual movement from centralised systems toward increasingly personal forms of computation.

Personal computers transferred computational capability from institutional facilities to individuals. Local storage replaced dependence upon shared mainframes. Users acquired increasing authority over the software, documents and computational environments through which they worked.

The emergence of cloud-based cognitive systems introduces the possibility of a different trajectory.

Although interfaces appear personal and highly interactive, the underlying architecture frequently remains concentrated within remote infrastructures. Computational reasoning occurs elsewhere. Long-term context is maintained elsewhere. Architectural decisions are made elsewhere. The user interacts through carefully designed interfaces while the structures that govern persistence, retrieval, configuration and continuity remain largely invisible.

In this respect, contemporary cognitive services may reproduce a relationship that resembles, in important architectural respects, earlier models of centralised computing.

Intelligence becomes concentrated within remote infrastructures.

- Users access capabilities rather than architectures.
- Persistence depends upon provider decisions.
- Access remains conditional.
- Operational policies determine acceptable forms of use.

The accumulated cognitive history developed through years of interaction may gradually become inseparable from the service through which it was created.

This paper does not suggest that distributed services are inherently undesirable, nor that remote computation lacks important advantages. Large-scale infrastructure will continue to play an essential role in many forms of cognitive computing.

The concern is architectural rather than geographical.

When continuity, memory, governance and operational authority become inseparable from external infrastructures, the structural relationship between individual and computing begins to resemble a form of cognitive recentralization.

The historical paper identified this possibility as one of the defining risks accompanying the transition toward cognitive computing.

The issue is not whether remote intelligence exists.

The issue is whether the individual continues to possess an enduring cognitive architecture that remains fundamentally their own.

3.3 The Invisible Transfer of Authority

Human beings have always delegated work to tools and the introduction of cognitive systems does not alter that principle.

What changes is the nature of the work being delegated and, consequently, the forms of authority that may be transferred without being consciously recognised.

The most familiar form is **functional delegation**.

A person asks a system to summarise a report, translate a document, analyse evidence or draft correspondence. The task is delegated while the surrounding environment remains substantially under the user's control. The system performs a function on behalf of the individual.

A second form is **architectural delegation**.

Here, the delegated activity extends beyond the task itself. The architecture through which cognition is organised becomes governed by another party. Memory, retrieval, persistence, operational history, identity management and the conditions under which the environment continues to exist are no longer primarily determined by the individual. Although the user may continue to perform the same functional tasks, authority over the surrounding cognitive environment has shifted elsewhere.

A third form is **cognitive delegation**.

Rather than merely executing discrete operations, computational systems begin to organise information, identify relevant evidence, establish relationships between documents, propose interpretations and shape the pathways through which understanding develops. Properly engineered, such assistance can significantly extend human capability. Improperly understood, it can obscure where interpretation originates and how conclusions emerge.

These distinctions are essential because they clarify a principle that remains constant throughout Personal Cognitive Computing.

Delegation is not abdication, and individuals may legitimately delegate computational work. They may choose to employ local models, remote services, specialised reasoning systems or collaborative infrastructures according to their own needs.

None of these choices transfers the ultimate responsibility for meaning, judgement or consequence.

- Interpretation remains a human responsibility.
- Decisions remain human decisions.
- Consequences remain human consequences.

The engineering objective of Personal Cognitive Computing is therefore not to eliminate delegation, but to ensure that delegation occurs within architectures that preserve transparency, continuity and meaningful human authority.

Only then can cognitive systems extend human reasoning without silently assuming control over the environments in which that reasoning takes place.

4. Why Local AI Does Not Solve the Problem

4.1 Local Execution Is Only One Property

The increasing availability of powerful open-weight language models has transformed the practical landscape of artificial intelligence. Individuals can now execute sophisticated reasoning systems on consumer hardware that, only a short time ago, required dedicated data centres or commercial cloud platforms. This development represents an important milestone in the democratisation of computational capability.

It does not, however, resolve the engineering problem defined by Personal Cognitive Computing.

The distinction is easily overlooked because local execution is both visible and measurable. Whether inference occurs on personal hardware or within a remote data centre is a clear technical characteristic. It influences performance, privacy, operating costs and availability.

Yet architecture extends beyond inference.

A language model may execute entirely on a user's own computer while the surrounding cognitive environment remains dependent upon external infrastructures. Long-term memory may continue to reside within provider-controlled systems. Identity and authentication may remain tied to cloud services. Documents may be synchronised through proprietary platforms. Retrieval may depend upon external indexing services. Telemetry may remain opaque. Application orchestration may occur through hosted infrastructures. The operational state accumulated through months or years of interaction may remain difficult to inspect, migrate or preserve independently.

In such circumstances, inference has become local while the architecture has not.

The user possesses computational capability but not necessarily architectural authority.

This distinction defines one of the central principles of Personal Cognitive Computing:

Local inference is a technical property. Personal Cognitive Computing is an architectural property.

The location at which computation occurs is important, but it is not sufficient. The defining question remains whether the individual's cognitive environment can exist as a durable, inspectable and user-governed system whose continuity does not depend upon the assumptions, policies or commercial interests of any particular implementation.

A system may satisfy every technical criterion for local execution while failing entirely to establish such an environment. Conversely, particular architectural components may legitimately be distributed across external infrastructures, provided the governing cognitive environment remains organised around the individual rather than the service.

The engineering objective is therefore not simply to relocate computation: it is to establish enduring architectural authority.

4.2 Model Ownership Is Not Architectural Ownership

The widespread availability of open-weight language models has understandably encouraged the belief that ownership of models is equivalent to ownership of cognitive systems.

The two are fundamentally different:

A language model is a reasoning component, it transforms inputs into outputs according to learned statistical representations.

Its weights determine how reasoning is performed, but they do not determine how evidence is governed, how knowledge accumulates, how instructions are organised or how cognitive continuity is preserved.

Owning model weights therefore answers only one architectural question:

- It does not establish who governs the evidence upon which reasoning operates.
- It does not define how long-term memory is organised or preserved.
- It does not determine the hierarchy through which user instructions, retrieved evidence and system behaviour interact.
- It does not provide traceability between conclusions and supporting sources.
- It does not ensure continuity of knowledge across changing models, applications or services.
- It does not establish user-controlled scope, nor does it guarantee that the accumulated cognitive environment can be migrated, reconstructed or recovered independently of the software through which it was created.

These concerns exist at a different architectural level.

They concern the environment surrounding reasoning rather than the reasoning mechanism itself.

Therefore, Personal Cognitive Computing deliberately separates the enduring cognitive environment from the replaceable reasoning systems that operate within it.

Models improve, architectures should endure.

4.3 Capability Does Not Equal Agency

Advances in artificial intelligence are commonly measured in terms of capability.

Benchmarks compare reasoning performance, factual accuracy, coding ability, mathematical competence and the growing range of tasks that computational systems can successfully perform. These measurements are valuable because they describe what a system is capable of computing.

They do not, however, describe who governs the architecture through which those computations occur.

A highly capable reasoning system may substantially increase the range of problems that an individual can solve while leaving unchanged the user's practical authority over memory, retrieval, persistence, evidence selection, operating history or long-term continuity.

Capability has increased where agency has not.

This distinction becomes increasingly important as cognitive systems assume more significant roles within research, professional work and personal decision-making.

A system that produces excellent answers is not necessarily a system that preserves user authority. Likewise, architectural agency should not be confused with computational performance.

A modest reasoning model operating within a transparent, inspectable and user-governed cognitive environment may preserve greater long-term authority than a substantially more capable model embedded within an architecture that remains opaque, non-portable or externally governed.

Personal Cognitive Computing therefore distinguishes between two fundamentally different dimensions of progress.

- **Capability** concerns what a model or computational system can compute.
- **Agency** concerns who owns, governs and can meaningfully inspect, preserve, challenge and ultimately control the architecture performing that computation.

The two dimensions are complementary rather than contradictory.

The objective of Personal Cognitive Computing is not to sacrifice capability in pursuit of agency, nor to reject advances in artificial intelligence. It is to ensure that increasing computational capability is accompanied by a corresponding preservation of human architectural authority.

Only when both dimensions advance together can cognitive systems genuinely extend human capability without diminishing the individual's enduring control over the environment in which that capability is exercised.

5. The Hardware Accessibility Problem

5.1 Formal Availability versus Practical Accessibility

The growing availability of open-weight language models is frequently presented as evidence that advanced artificial intelligence has become accessible to everyone.

This conclusion deserves closer examination.

A model may be publicly downloadable while remaining practically unavailable to most individuals. Formal access does not necessarily imply meaningful access. A model that requires hardware far beyond the economic reach of ordinary users, consumes impractical amounts of energy or performs inadequately on commonly available systems cannot reasonably be regarded as broadly accessible, regardless of its licensing terms.

This distinction introduces several different forms of availability.

Theoretical access concerns whether a model can be obtained at all.

- **Economically** attainable access concerns whether individuals can realistically acquire the hardware necessary to operate it.
- **Technically** usable access concerns whether that hardware can execute the model with sufficient performance to support everyday cognitive work.
- **Socially** meaningful ownership concerns whether these capabilities become available to society at large rather than remaining concentrated within a relatively small group of organisations or individuals possessing exceptional computational resources.

Personal Cognitive Computing is concerned with the last of these.

The objective is not merely that cognitive capabilities exist, but that they become practically available as durable personal infrastructure for ordinary people.

5.2 Consumer Hardware as a Defining Constraint

For this reason, affordable consumer-grade hardware is not treated as an implementation preference or a later optimisation.

It constitutes one of the defining engineering constraints of the discipline.

Architectures that assume unlimited computational resources inevitably produce solutions whose practical ownership remains restricted. By contrast, architectures designed explicitly around realistic personal hardware must confront the physical and economic limitations experienced by the overwhelming majority of users.

These limitations are neither temporary inconveniences nor secondary engineering concerns.

Available memory constrains the scale of models that can be executed. Inference speed determines whether cognitive assistance remains practical during everyday work. Storage capacity influences the long-term preservation of personal knowledge. Energy consumption and thermal behaviour affect operational viability outside specialised computing environments. System cost determines who can realistically participate, while maintenance, repairability and expected hardware lifespan influence whether cognitive infrastructure remains sustainable over many years of continuous use.

These constraints therefore shape architectural decisions from the outset: rather than designing systems that require exceptional hardware and subsequently attempting to reduce their resource demands, Personal Cognitive Computing begins with the assumption that meaningful cognitive ownership should be achievable on hardware that ordinary individuals can reasonably obtain, operate and maintain.

The engineering discipline must therefore optimize not only for computational capability, but also for practical accessibility.

5.3 Why This Matters Beyond Product Design

The accessibility of hardware extends beyond questions of product engineering or consumer preference, it directly influences the distribution of cognitive authority within society.

If meaningful ownership of advanced cognitive environments remains practical only for governments, large corporations, specialised laboratories or wealthy individuals, then the widespread availability of portable reasoning models does not fundamentally alter the underlying distribution of power. Computational capability may have become technically portable, while meaningful cognitive ownership remains economically concentrated.

Such an outcome would fall short of the central objective established throughout these papers.

Personal Cognitive Computing seeks to establish architectures through which individuals can exercise enduring authority over their own cognitive environments. That objective cannot be realised if the necessary infrastructure remains inaccessible to the vast majority of those whom the discipline is intended to serve.

Hardware accessibility therefore represents more than a question of engineering efficiency and it is a public-interest concern.

The affordability, maintainability and longevity of personal computing platforms determine whether Personal Cognitive Computing becomes a broadly attainable foundation for cognitive sovereignty or remains a capability available only to those with exceptional resources.

For that reason, engineering for consumer-grade hardware is not simply about reducing costs. It is about ensuring that the architectural principles of Personal Cognitive Computing remain achievable in practice for the ordinary individuals whose cognitive environments the discipline exists to protect.

6. The Evidence Problem

6.1 Language Models Are Not Sufficient Knowledge Authorities

The remarkable capabilities of modern language models have understandably encouraged their use as increasingly general-purpose knowledge systems. They explain, summarise, compare, infer and reason across an extraordinary range of subjects. These capabilities are central to the emergence of Personal Cognitive Computing.

They do not, however, make language models authoritative repositories of personal truth.

Personal Cognitive Computing deliberately assigns a different role to the model.

A language model is primarily a reasoning engine. Its purpose is to analyse, synthesise and interpret information. While it contains extensive statistical knowledge acquired during training, it cannot be presumed to possess authoritative knowledge about an individual's documents, history, circumstances or evolving body of work.

Whenever a consequential conclusion depends upon a person's own information, the foundation should therefore be explicit evidence rather than implicit model knowledge.

- The model contributes reasoning.
- The evidence contributes authority.

This distinction preserves a clear separation between what the model can infer statistically and what can be established from the user's own cognitive environment.

6.2 Retrieval and Reasoning Must Remain Distinct

This separation gives rise to one of the central engineering challenges of Personal Cognitive Computing.

Reasoning should operate upon explicitly constructed evidence rather than silently determining its own informational foundation.

Accordingly, retrieval and reasoning fulfil fundamentally different architectural responsibilities.

Retrieval identifies potentially relevant information within the Personal Cognitive Environment. Evidence construction transforms that retrieved information into an explicit and inspectable body of supporting material. Reasoning then operates upon that evidence to produce interpretation, explanation or recommendation. Finally, the resulting conclusions should remain traceable to the sources upon which they depend.

Each stage performs a distinct function. Collapsing these responsibilities into a single opaque process weakens transparency and makes meaningful inspection increasingly difficult.

For the same reason, reasoning should not silently expand its scope beyond the evidence that has been deliberately established. Hidden searches, undisclosed retrieval operations or unobservable expansion of context undermine the user's ability to understand how conclusions have been reached and whether alternative evidence might reasonably have produced different outcomes.

The objective is therefore not to constrain reasoning, but to ensure that its informational foundation remains explicit, inspectable and governed by the architecture rather than by hidden computational behaviour.

6.3 Evidence Selection Is Itself Consequential

The quality of a cognitive system cannot be evaluated solely by examining the sophistication of its reasoning. Its conclusions depend equally upon the evidence from which that reasoning proceeds. Every stage preceding inference therefore carries epistemic significance.

What information was successfully indexed determines what can subsequently be discovered. Retrieval determines which material is considered relevant. Excluded information may prove just as consequential as information that is included. The manner in which documents are segmented influences the context available to reasoning. Query formulation shapes the search itself. Ranking mechanisms determine which sources receive attention, while the final evidence set establishes the informational foundation upon which every subsequent conclusion rests.

Reasoning cannot compensate for evidence that was never presented, nor can transparent conclusions emerge from opaque evidence selection.

For this reason, Personal Cognitive Computing treats retrieval as substantially more than an internal implementation detail. It forms an integral part of the epistemic architecture through which knowledge is constructed. Decisions made during indexing, retrieval and evidence formation directly influence the conclusions a system is capable of reaching.

Engineering trustworthy cognitive environments therefore requires transparency not only in reasoning, but also in the processes through which evidence is discovered, selected and assembled. Only when both remain inspectable can users meaningfully understand, evaluate and, where necessary, challenge the foundations of computational reasoning.

7. The Memory Problem

7.1 Persistence Creates Power

One of the defining characteristics of cognitive systems is their ability to remember.

Unlike traditional software, which often treats each interaction as an isolated event, cognitive systems increasingly accumulate knowledge across conversations, projects and extended periods of use. This persistence enables continuity, reduces repetition and allows computational reasoning to build upon previous work. Properly governed, persistent memory becomes one of the most valuable capabilities within a Personal Cognitive Environment.

It also creates one of its greatest engineering challenges.

Memory is never a neutral technical feature. The ability to preserve information implies the ability to influence future reasoning. What is remembered may shape subsequent interpretation. What is forgotten may disappear from future consideration. Persistence therefore becomes a source of architectural authority.

The central question is consequently not how memory can be stored, the more fundamental question is under what authority information may become persistent, remain active, be modified, be challenged and ultimately be forgotten.

These decisions cannot be treated as incidental implementation details. They determine how cognitive environments evolve over time and who retains meaningful control over that evolution.

Personal Cognitive Computing therefore regards memory not merely as stored information, but as a governed architectural responsibility.

7.2 Memory Must Be Governed

The tendency to describe all retained information simply as "memory" conceals important architectural distinctions. Not all information serves the same purpose, nor should it be governed according to the same rules.

Temporary interaction context exists only to support the immediate exchange between user and system. Session state maintains continuity within a particular working session. Project state preserves information relevant to a specific body of work without necessarily becoming part of an individual's enduring cognitive environment.

Beyond these transient forms lie durable categories whose significance extends much further. Personal memory reflects information intentionally retained across time. Source documents remain authoritative records that should not be confused with derived interpretations. Inferred memories arise through computational analysis and therefore require different governance from explicitly authored knowledge.

User-authored instructions express intentional guidance for future system behaviour, while system-generated summaries represent interpretations that remain open to revision rather than becoming unquestioned fact.

These categories perform fundamentally different architectural roles.

Collapsing them into a single undifferentiated context store obscures provenance, weakens governance and makes it increasingly difficult for users to understand why information persists or how it influences future reasoning.

Personal Cognitive Computing therefore requires explicit distinctions between different forms of persistence.

Governance begins with recognising that not every form of memory should be treated as though it were the same.

7.3 Memory Must Remain Inspectable and Reversible

Persistence without transparency inevitably concentrates authority within the system rather than the individual.

A user should therefore never be required to speculate about the operation of memory.

It should remain possible to determine what information has been retained, why it was considered worthy of preservation, where it originated, when it entered the cognitive environment and how it contributes to subsequent reasoning. Equally important, the user should be able to challenge incorrect information, revise outdated interpretations and remove knowledge that no longer belongs within the enduring cognitive environment.

These capabilities are not conveniences, they are prerequisites for meaningful cognitive ownership.

A memory that cannot be inspected becomes difficult to trust. A memory that cannot be corrected becomes difficult to govern. A memory that cannot be forgotten gradually ceases to belong to the individual whose cognitive environment it purports to serve.

For this reason, the canonical architecture assigns persistent memory to the governed responsibilities of the Personal Cognitive Core. Memory is not conceived as a passive storage mechanism, but as an architectural function operating under explicit user authority. Its purpose is to preserve continuity without diminishing agency, ensuring that persistence strengthens the Personal Cognitive Environment while remaining permanently subject to human inspection, correction and control.

8. The Continuity Problem

8.1 Models Are Temporary; Cognitive Environments Endure

Technological change has always been a defining characteristic of computing.

Programming languages evolve, operating systems are replaced, hardware platforms become obsolete and software frameworks rise and disappear within remarkably short periods of time. Artificial intelligence has accelerated this pattern still further. Language models that represent the state of the art today may be surpassed within months, while providers, licensing conditions and deployment strategies continue to change at an unprecedented pace.

Personal Cognitive Computing accepts this reality as a permanent condition rather than a temporary phase.

The engineering mistake would be to anchor a person's cognitive environment to the most rapidly changing elements of the computing ecosystem:

The accumulated knowledge of an individual often extends across decades rather than product cycles. Documents, correspondence, research, professional expertise, institutional knowledge and carefully developed interpretations frequently outlive the software through which they were first created. Their value lies precisely in their continuity.

The enduring object of engineering should therefore not be the reasoning model, it should be the cognitive environment that remains meaningful as successive generations of computational components are replaced. Models are temporary, personal cognitive environments are intended to endure.

8.2 Epistemic Continuity

For this reason, Personal Cognitive Computing identifies **epistemic continuity** as one of its central engineering objectives.

Epistemic continuity is the ability to preserve and reconstruct the evolving body of knowledge upon which an individual or organisation depends, despite changes in technology, infrastructure or institutional context.

This extends well beyond the preservation of documents alone.

A durable cognitive environment must retain not only information, but also its provenance, the instructions that governed computational behaviour, the evidential relationships connecting sources to conclusions, the processing histories through which results were produced, the interpretations developed over time, the decisions that followed from those interpretations and the institutional memory that gives accumulated knowledge its practical significance. Together, these elements allow knowledge to remain intelligible rather than merely recoverable.

An archived document without its evidential context, processing history or surrounding interpretations may preserve information while losing understanding. Epistemic continuity therefore concerns the preservation of meaning as well as data.

Operationally, it requires that cognitive environments remain coherent across changes in software platforms, hardware generations, reasoning models, service providers and organisational structures.

Only then can computational progress strengthen long-term knowledge rather than repeatedly fragmenting it.

8.3 Recoverability

Continuity cannot exist without recoverability. A cognitive environment whose continued existence depends upon the survival, policies or commercial interests of a single provider cannot be regarded as fully sovereign, regardless of its computational capabilities.

Recoverability therefore becomes an architectural requirement rather than an operational convenience.

A Personal Cognitive Environment should remain exportable in transparent formats that preserve its meaning beyond a particular application. It should support reliable backup and restoration without hidden dependencies. Migration to new hardware, software platforms or reasoning systems should preserve continuity rather than requiring reconstruction from scratch. Computational components—including language models, retrieval engines and user interfaces—should remain replaceable without compromising the integrity of the underlying cognitive environment. Equally important, consequential outputs should remain reconstructable together with the evidence upon which they were based, allowing previous reasoning to be understood, verified and, where necessary, challenged.

These capabilities collectively define the practical meaning of recoverability.

Their purpose is not merely to protect against technical failure; they ensure that an individual's accumulated cognitive environment remains fundamentally independent of any single implementation, provider or generation of technology.

In this way, recoverability becomes the operational expression of epistemic continuity.

It enables knowledge to survive technological change while preserving the user's enduring authority over the cognitive environment through which that knowledge is created, maintained and applied.

9. The Governance Problem

9.1 Governance as Architecture

As cognitive environments become increasingly capable, governance can no longer be regarded as an external policy layer applied after a system has been built.

Governance is itself an architectural responsibility.

Every cognitive system embodies decisions about authority. It determines what information enters the environment, what information may subsequently be retrieved, what becomes persistent, which instructions take precedence when multiple directives exist, which computational models are permitted to operate, how external services may participate, which actions require explicit confirmation and what forms of activity become part of the permanent operational record.

These are not administrative concerns; they define the behaviour of the architecture itself.

Whether explicit or implicit, every implementation already answers such questions. The engineering challenge is therefore not whether governance exists, but whether it remains visible, understandable and ultimately subject to the authority of the individual whose cognitive environment is being governed.

Personal Cognitive Computing requires governance to be designed as an inspectable architectural capability rather than embedded invisibly within application logic or provider policy. Only then can users understand not merely what a system does, but under which rules it operates.

9.2 Advice versus Decision

As cognitive systems become more capable, they increasingly assist users by identifying relevant evidence, evaluating alternatives, ranking possible actions and highlighting potential risks. These capabilities represent an important extension of human reasoning and form a legitimate part of Personal Cognitive Computing.

They do not, however, alter the location of final authority.

Whenever multiple reasonable courses of action exist, the system may analyse them, compare them and explain their respective consequences. It may recommend one approach over another or warn against decisions that appear inconsistent with available evidence.

The final decision nevertheless remains with the individual.

This distinction is not merely ethical, it is an engineering principle that determines how cognitive systems should be designed.

A system organised around advice preserves human judgement while extending human capability. A system organised around autonomous decision-making gradually transfers authority from the user to the architecture itself.

For this reason, Personal Cognitive Computing adopts a foundational rule:

The system may advise. The user decides.

This principle does not diminish the value of computational reasoning, it further defines the relationship within which that reasoning is exercised.

9.3 Bounded Delegation

The preservation of cognitive sovereignty should not be mistaken for technological isolation.

Personal Cognitive Computing does not require that every computation occur locally, nor does it prohibit collaboration with external models, specialised services or distributed infrastructures. Such capabilities may offer significant advantages and will remain an important part of future cognitive ecosystems.

The engineering question is therefore not whether delegation occurs; it is how delegation is governed.

Delegation should always be explicit rather than assumed. Its scope should remain clearly bound to the purpose for which it was authorised. Its operation should be visible to the user rather than concealed within opaque processing pipelines. Its authority should remain limited to the agreed function without silently expanding into unrelated responsibilities. Its outcomes should remain open to inspection and challenge, and the delegated relationship itself should be reversible without compromising the continuity of the Personal Cognitive Environment.

These properties establish delegation as a controlled architectural mechanism rather than an open-ended transfer of authority.

They ensure that external services extend the capabilities of the cognitive environment without becoming its governing centre.

In this way, Personal Cognitive Computing distinguishes sovereignty from isolation.

The objective is not to eliminate collaboration with external computational resources, but to ensure that such collaboration always occurs within architectures whose enduring authority remains with the individual rather than with the services upon which they temporarily rely.

10. The Observability Problem

10.1 Prompt and Answer Are Insufficient

Most contemporary conversational systems present cognition as a black box.

The user provides a prompt. After a period of computation, an answer appears. Everything that occurred between those two moments remains largely invisible.

For many applications this level of abstraction is entirely appropriate. Users need not understand the internal operation of every computational process they employ.

Personal Cognitive Computing, however, concerns systems that increasingly participate in research, professional work, knowledge management and consequential decision-making. In such environments, the process by which an answer is produced often matters as much as the answer itself: a prompt and a response alone are therefore insufficient.

Without visibility into the intermediate stages of processing, the user cannot readily determine whether relevant information was available, whether retrieval succeeded, whether evidence was incomplete or whether a failure occurred before reasoning even began.

Opacity not only limits understanding, it also limits trust, because conclusions cannot be meaningfully evaluated when the processes that produced them remain concealed.

Observability is therefore not introduced to satisfy technical curiosity, it exists to make consequential cognitive processes understandable.

10.2 Observable Stages

The objective of observability is not to expose every internal implementation detail or every computational operation performed by the system.

Rather, it is to make the significant architectural stages visible wherever they contribute to the user's understanding of how a result was produced.

A Personal Cognitive Environment should therefore make it possible, where appropriate, to observe the progression of work through the cognitive pipeline. The user should be able to determine that a request has been received, whether intent clarification has taken place, whether indexing is complete, which retrieval mode has been employed, how retrieval queries were formulated, which sources were identified, how evidence was assembled, when responsibility passed to the reasoning model, whether verification occurred, whether processing completed successfully or failed, and how much time each stage required.

These stages represent meaningful transitions in the construction of an answer.

They reveal not the internal mathematics of a language model, but the architectural pathway through which information became reasoning.

Such visibility enables users to distinguish between retrieval failures and reasoning failures, between missing evidence and insufficient interpretation, and between successful completion and interrupted execution.

Observability therefore transforms computational processing from an opaque event into an inspectable sequence of architectural operations.

10.3 Transparency Without Overwhelming the User

Observability should not be confused with unrestricted technical exposure.

An architecture that presents every internal operation risks replacing opacity with incomprehensibility. Excessive detail can obscure rather than illuminate the processes that matter most to users. The objective is therefore selective transparency.

The architecture should provide sufficient visibility for users to understand what occurred, which information contributed to the result, where failures or interruptions arose and which aspects of the outcome remain uncertain or incomplete. It should explain the progression of cognitive processing without requiring expertise in implementation details or model internals.

This balance reflects a broader principle within Personal Cognitive Computing.

Transparency exists to strengthen human understanding, not to maximise technical disclosure.

The appropriate level of observability is therefore determined by what enables meaningful inspection, informed judgement and effective governance of the Personal Cognitive Environment.

Within the current reference implementation, Nivel's Trace Log and explicit evidence display represent early experiments in this direction. They do not attempt to expose every internal computation. Instead, they seek to make the significant stages of cognitive processing visible, allowing users to follow the architectural path from request to evidence, from evidence to reasoning and from reasoning to conclusion.

In this sense, observability is not an auxiliary feature added for debugging. It is a fundamental architectural property through which cognitive systems become inspectable, understandable and ultimately worthy of trust.

11. Formal Statement of the Engineering Problem

The preceding chapters have argued that the emergence of Personal Cognitive Computing is not driven by the invention of increasingly capable language models alone. It arises from a more fundamental architectural transition.

As computational systems become participants in human reasoning, research, memory and decision-making, the primary engineering object is no longer the isolated application, the individual model or the computational service. It is the enduring cognitive environment within which these components operate and through which human knowledge is accumulated, organised and preserved. This transition gives rise to a distinct engineering problem.

The problem cannot be reduced to improving model capability, increasing computational performance or enabling local execution. Nor can it be solved by open-source licensing, stronger privacy guarantees or more sophisticated retrieval mechanisms in isolation.

Each addresses an important aspect of cognitive computing.

None, individually, resolves the architectural challenge that defines the discipline.

The engineering problem of Personal Cognitive Computing is therefore formulated as follows:

The engineering problem of Personal Cognitive Computing is to design cognitive environments in which individuals and communities can retain meaningful authority over their information, memory, evidence, instructions, computational services and accumulated knowledge, while obtaining useful cognitive assistance on affordable and realistically maintainable infrastructure.

This formulation deliberately places authority alongside capability.

The objective is not merely to construct increasingly intelligent computational systems; it is to ensure that increasing intelligence is accompanied by architectures that preserve continuity, inspectability, recoverability and meaningful human governance.

Achieving this objective requires the reconciliation of a series of enduring engineering tensions rather than the optimisation of any single property.

Greater computational capability must remain compatible with practical affordability. Increasing automation must not diminish meaningful user control. Personalisation should strengthen utility without undermining privacy. Persistent memory must preserve continuity while remaining open to correction and forgetting. Immediate convenience should not compromise long-term continuity. Openness must be balanced with security. External computational services should extend cognitive capability without displacing architectural authority. Higher levels of abstraction should not eliminate inspectability. Improvements in performance should never render evidence or reasoning impossible to reconstruct.

These tensions are not temporary implementation challenges.

They represent the continuing research agenda of the discipline itself.

No single architecture will permanently resolve them, and no technological advance will eliminate the need to balance them. As computational capabilities evolve, these relationships must be continually re-evaluated in light of changing hardware, software, organisational structures and societal expectations.

For this reason, the engineering problem of Personal Cognitive Computing cannot be solved by any single language model, software product, deployment topology or commercial platform.

It is solved, progressively and collectively, through the development of architectures that preserve enduring human authority while embracing continual technological change.

That objective defines both the purpose of the discipline and the research programme that follows.

Chapter II - The Research Agenda for Personal Cognitive Computing

12. Purpose of the Research Agenda

The preceding chapter defined the engineering problem that gives rise to Personal Cognitive Computing as a distinct engineering discipline. It argued that the fundamental challenge is not merely the development of increasingly capable language models, but the construction of cognitive environments through which individuals and communities can retain meaningful authority over their information, memory, evidence, instructions, computational services and accumulated knowledge while benefiting from computational reasoning.

Defining the problem, however, is only the first step in establishing a discipline.

A discipline matures through sustained investigation. It develops shared concepts, repeatable methods, experimental evidence, reference architectures and progressively more robust engineering principles. These do not emerge from a single product or isolated implementation. They require a coherent programme of research extending across technologies, architectures, applications and successive generations of computational capability.

The purpose of this chapter is to define that programme.

It identifies the enduring research domains that Personal Cognitive Computing must investigate if it is to develop into a rigorous and practically consequential field of engineering. Together, these domains establish the intellectual territory of the discipline: the principal problems that remain unresolved, the relationships between them and the areas in which systematic experimentation, evaluation and technical development are required.

The agenda is deliberately independent of any particular implementation.

Language models will improve. Hardware capabilities will change. Software frameworks will be replaced, and new computational paradigms will emerge. The underlying questions of cognitive authority, evidence, memory, governance, continuity, inspectability and accessibility will nevertheless remain. The research programme must therefore be durable enough to guide investigation beyond the assumptions and technical limitations of the present moment.

At this foundational stage, the objective is not to answer every research question or prescribe detailed experimental programmes for every domain. Nor is it to present a development roadmap for a particular system. Its purpose is to establish the complete field of inquiry within which future research can be organised.

Each domain may subsequently be developed into a more detailed research programme defining its specific problem, principal questions, experimental approaches, expected outputs, relationship to the mission, current maturity and unresolved tensions. Those specifications should emerge progressively as the field develops and as sufficient evidence exists to formulate them responsibly.

The research agenda also provides an institutional framework for sustained work.

It allows research projects to be selected and coordinated as contributions to a common scientific purpose rather than as disconnected technical experiments. It provides a basis for collaboration with researchers, engineers, universities, civil-society organisations and other institutions whose work intersects with the emerging field. It also makes visible where financial support, infrastructure, specialist knowledge and long-term investigation are required.

This is especially important because the central problems of Personal Cognitive Computing cannot be resolved through short-term product development alone. Questions concerning persistent memory, cognitive continuity, evidence-centred reasoning, architectural governance, affordable infrastructure and consequential use require longitudinal research, comparative experimentation, reference implementations, evaluation methods and publicly examinable results.

The agenda therefore serves both a scientific and an institutional purpose.

Scientifically, it establishes the enduring questions through which the discipline can advance. **Institutionally**, it provides the Institute with a coherent basis upon which to organise research, form partnerships, seek funding and demonstrate how individual projects contribute to a larger public mission.

Funding such a programme is not simply funding another software platform. It is supporting the development of concepts, methods, architectures and evidence needed to ensure that computational reasoning can become durable human infrastructure without placing personal and collective cognitive life under the permanent authority of a small number of technological providers.

The objective of this chapter is therefore not to resolve the research agenda.

It is to establish it: to define a stable foundation upon which future research, experimentation, collaboration and engineering advances can progressively build.

13. Research Domain I — Personal Cognitive Environments

This domain investigates what constitutes a Personal Cognitive Environment as a durable technical object.

It concerns the representation, organisation, provenance, boundaries, portability and lifecycle of the information, instructions, histories and accumulated knowledge upon which cognitive applications operate.

14. Research Domain II — Scope and Intent

This domain investigates how the intended boundary of a cognitive operation can be made explicit before retrieval or reasoning begins.

It concerns intent clarification, scope representation, user confirmation and the preservation of the distinction between the user's original request and any computationally optimised formulation of that request.

15. Research Domain III — Retrieval Intelligence

This domain investigates how relevant information can be identified effectively while keeping retrieval visible, understandable and subject to user control.

It includes lexical, semantic, metadata-driven, hybrid and document-aware retrieval, as well as retrieval modes, retrieval advisories, uncertainty, duplication and failure diagnosis.

16. Research Domain IV — Evidence Construction

This domain investigates how retrieved material becomes an explicit, bounded and traceable evidence set.

It concerns the units of evidence, contextual completeness, provenance, source authority, contradiction, duplication and the user's ability to inspect or modify the evidence available for reasoning.

17. Research Domain V — Evidence-Centred Reasoning

This domain investigates how reasoning systems can operate within explicit evidential and instructional boundaries.

It concerns the distinction between source material and inference, the treatment of insufficient or conflicting evidence, citation, uncertainty and the appropriate role of model-internal knowledge.

18. Research Domain VI — Verification

This domain investigates what it means to verify the outputs of a personal cognitive system.

It concerns claim-to-evidence relationships, citation integrity, unsupported assertions, conflicting sources, confidence, automated checking and the continuing role of human review.

19. Research Domain VII — Persistent Memory

This domain investigates how persistent memory can improve continuity without becoming invisible, uncontrollable or authoritative in its own right.

It concerns memory creation, consent, provenance, inference, correction, contradiction, versioning, expiration, deletion, portability and the visibility of memory's influence on subsequent reasoning.

20. Research Domain VIII — Instruction Governance

This domain investigates how instructions should be represented, prioritised, applied and inspected.

It concerns constitutional, user, project, workflow and temporary instructions; conflicts between them; their influence on cognitive operations; and their preservation across applications and systems.

21. Research Domain IX — Traceability and Observability

This domain investigates what users must be able to observe in order to understand how a cognitive result was produced.

It concerns execution stages, retrieval and model handoffs, failures, retries, timing, system configuration, audit records and the distinction between observable system behaviour and inaccessible model internals.

22. Research Domain X — Architectural Governance

This domain investigates how authority is distributed between the user, the cognitive environment, applications, automated processes and external services.

It concerns permissions, confirmation, delegation, sensitive information, shared stewardship, multi-user environments and the prevention of computational capability from silently becoming architectural authority.

23. Research Domain XI — Cognitive Continuity

This domain investigates how cognitive environments can remain coherent across technological and institutional change.

It concerns durable formats, canonical and derived information, migration, recovery, reconstruction, component replacement and the long-term preservation of provenance, relationships, interpretations and institutional knowledge.

24. Research Domain XII — Local Cognitive Services

This domain investigates how language models, embedding systems, retrieval engines, parsers, optical-character-recognition systems and other computational services can remain replaceable components within a stable architecture.

It concerns service interfaces, capability declarations, resource requirements, failure behaviour, model replacement and governed interaction between local and external services.

25. Research Domain XIII — Affordable Hardware and Systems Engineering

This domain investigates what useful cognitive capability can be delivered on hardware ordinary people can realistically own, operate and maintain.

It concerns model and context requirements, memory use, inference performance, storage, energy, thermal behaviour, cost, repairability, upgradeability and graceful operation under constrained resources.

26. Research Domain XIV — Human-System Interaction

This domain investigates how sophisticated cognitive infrastructure can remain understandable and governable without requiring every user to become a systems engineer.

It concerns transparency, uncertainty, meaningful control, evidence inspection, memory governance, accessibility and the design of interfaces appropriate to different levels of technical understanding.

27. Research Domain XV — Domain Solutions and Consequential Use

This domain investigates how the general architecture of Personal Cognitive Computing can support consequential fields without embedding their specialised assumptions within the common core.

Initial areas may include medical evidence, legal and administrative processes, scientific research, education, engineering knowledge and the continuity of small organisations.

The domain concerns the transparent incorporation of specialised evidence standards, professional limitations, domain knowledge, source authority and human responsibility.

28. Research Methodology

Personal Cognitive Computing requires an empirical research method through which architectural principles and specifications can be tested without being treated as permanently fixed.

Research within Personal Cognitive Computing follows the engineering methodology established in *Personal Cognitive Engineering*. The canonical progression is:

**Observation → Engineering Principle → Architectural Concept →
Architectural Specification → Shared Cognitive Capability → Reference
Implementation → Module Integration → Domain Solution**

Observations are translated into engineering principles, architectural concepts and specifications. These guide the development of shared cognitive capabilities, reference implementations, module integrations and, ultimately, domain solutions.

This progression is not intended as a one-directional development process. Within it, reference implementations, application experiments, module integrations and domain solutions generate evidence through evaluation. That evidence may validate, reject or refine shared capabilities, architectural specifications, architectural concepts and, where necessary, the underlying engineering principles.

Experimental capabilities should remain application-specific until evidence demonstrates that they are useful beyond a single application. Promotion into shared cognitive infrastructure should occur only after their usefulness, boundaries, observability, resource costs, reusability and compatibility with the mission of Personal Cognitive Computing have been sufficiently demonstrated.

Failed experiments, insufficient hardware, ineffective retrieval methods, unnecessary abstractions and forms of automation that reduce transparency are part of the research record. Negative findings are not peripheral to the programme. They help establish the practical limits of particular approaches, prevent unsuccessful assumptions from becoming embedded in the architecture and contribute to the cumulative development of the discipline.

Nivel serves as the present reference implementation through which experiments can be conducted and evidence gathered. It provides a practical environment for testing architectural ideas, shared capabilities and application-level mechanisms.

It does not determine the permanent architecture, implementation strategy or limits of Personal Cognitive Computing.

29. Research Outputs

The research programme may produce:

- constitutional principles;
- reference architectures;
- technical specifications;
- reference implementations;
- benchmark and hardware studies;
- design patterns;
- evaluation instruments;
- open datasets where appropriate;
- documented negative findings;
- educational materials;
- public demonstrations;
- domain prototypes.